Why the RelaLeap ePC Distillation Programme Failed, and a Redesigned Path to Cap-Ready ePC Transformers

Diagnostic analysis of the July–August 2026 experiment record, with an agent-executable remediation programme

Prepared for Ben Goertzel and the RelaLeap research-agent workflow

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Abstract

This report diagnoses the failed RelaLeap effort (15 July – 4 August 2026) to distill small transformers into error-based predictive-coding (ePC) networks, culminating in the multi-regime frontier preflight whose matched $T=1$ control failed its learnability gate in every regime. We argue that the programme's negative results are largely uninformative about ePC itself, for four reasons. (1) The implemented algorithm is state-based predictive coding (sPC), not the error-based reparameterization (EO/ePC) of Goemaere et al. (arXiv:2505.20137); the observed depth-decaying credit, representation-rank collapse, and contradiction with the parallel adjoint-field track are the predicted signatures of exactly this substitution. (2) The final preflight control scored *worse than chance* in every regime, which is an instrument defect (train/eval mapping mismatch or per-token vs. per-sequence NLL miscalibration), not a task difficulty result; no chance baseline was ever computed. (3) The programme's sole reproducible positive invariant—monotone activity energy—is true by construction of the backtracking line search and verifies nothing about convergence. (4) The repeated pursuit of a held-out-loss win over matched backpropagation contradicts the equilibrium equivalence between PC and BP weight gradients; matching BP is the theoretical ceiling of that comparison. We then reframe the objective around the stated goal—an ePC substrate whose internal representation is crisp, causal, and easy to build predictive-coding caps on top of—and specify a four-stage replacement programme (M0 analytic equivalence gates; C1 conversion-by-homotopy of competent transformers; C2 pre-registered cap-readiness metrics along the output-precision path; C3 a matched-cap falsifier). The appendix gives detailed, prescriptive guidance for the coding agents that will execute it, including a mandatory mechanism-gate tier with closed-form pass criteria.

Contents

1 Scope, sources, and summary

This analysis covers two documents of 4 August 2026—the failed multi-regime ePC frontier preflight report and its companion history of the prior ePC programme—together with the public reproduction bundle (`github.com/bgoertzel-sing/relaleap-epc-multiregime-preflight`, commit `eabae42`) and the reference method paper for error-based predictive coding, Goemaere, Oliviers, Bogacz, and Demeester, arXiv:2505.20137 ("Error Optimization," EO; here also "ePC").

The experiment record is procedurally excellent: frozen gates, disjoint exploration/selection/confirmation seed splits, bit-exact replay, SHA-256 digests of every artifact, and a 208-test suite. The problem is that this rigor was applied downstream of four upstream defects, so that the programme reproducibly certified the failure of a mislabeled algorithm against a miscalibrated gate on a broken fixture, while never running the one comparison its stated goal actually requires.

The four findings, in decreasing order of importance:

1. The implemented relaxation is *state-based* PC, not error-based PC. Section 2.
2. The final preflight control failure is an instrument bug signature, not a difficulty signature. Section 3.
3. Energy monotonicity, the programme's flagship invariant, is vacuous. Section 4.
4. The loss-improvement question the programme kept asking has a known theoretical answer ("no, by design"), and the substrate-quality question it should have asked was never operationalized. Section 5.

Sections 6–7 give the redesigned programme; Section 9 extracts the process lesson; the Appendix is written to be handed directly to coding agents.

2 Finding 1: the implemented algorithm is sPC, not ePC

2.1 What was implemented

Both reports declare the local energy

$$E(z) \;=\; \tfrac{1}{2}\sum_{l} \mathrm{mean}\big\|z_l - f_l(z_{l-1})\big\|^2 \;+\; \lambda\,\mathrm{KL}\big(q \,\|\, \mathrm{softmax}(\mathrm{logits}(z_L)/\tau)\big), \tag{1}$$

with relaxation performed by gradient steps *on the block states* z_l, accepted by a backtracking line search, for $T \in \{2, 4, 12\}$ recorded settlement evaluations.

2.2 What ePC actually is

Optimizing (1) over the states z is textbook state-based predictive coding (sPC). The entire contribution of arXiv:2505.20137 is a *reparameterization*: define the prediction errors

$$\epsilon_l \;\stackrel{\text{def}}{=}\; z_l - f_l(z_{l-1}), \tag{2}$$

substitute $z_l = f_l(z_{l-1}) + \epsilon_l$ recursively, and optimize the energy *over the errors* ϵ. Under the state parameterization, a change at the output must propagate to early layers through the chain of local quadratic couplings, and on digital hardware the inference signal attenuates geometrically with depth; the authors show this makes deep sPC pathologically slow or effectively untrainable, and they warn explicitly that experiments run with sPC lead to incorrect conclusions about predictive coding. Under the error parameterization, the output discrepancy acts on every layer's error variable directly in the first inference step; signal reaches all depths simultaneously and unattenuated.

Nothing in either report, or in the extracted source bundle, performs this reparameterization. The programme tested sPC while citing the ePC paper. (The public README even glosses ePC as "exploratory predictive coding.")

2.3 The recorded data are the predicted sPC signatures

This is not a labeling quibble; it retrodicts the record almost quantitatively.

- **Depth-decaying credit (15 July gate).** Before the "normalization fix," the block-1 ePC/KD gradient-norm ratio was $\sim 10^{-4}$ and block-0 credit was zero: geometric attenuation with depth, exactly the sPC pathology. The fix (raising the ratio to 0.050/0.136/0.265 at T=2/4/8) rescaled the symptom without removing the mechanism.

- **Shallow settlement.** At $T \leq 12$ steps on a six-layer network, sPC is nowhere near its energy fixed point. The "settled" state is a small perturbation of the feedforward pass, so the weight update is a slightly perturbed KD update. This predicts the R7/R8 observation that ePC arms land within seed noise of update-matched BP+KD while costing 6–45× the wall clock, and that T=12 does not improve on T=4.
- **Late-block energy domination and rank collapse (R8/R9).** With a single global output weight λ, temperature τ=1, and un-normalized residual-stream coordinates (whose norms grow with depth), the output KL term and the deepest quadratic terms dominate (1). Relaxation then drags the late-layer states onto the low-dimensional manifold that reproduces the teacher logits while early blocks barely move. That is compression *caused by decay*: effective rank 3.8–3.9 vs. 13.9 for BP+CE, with uniformly weaker causal selectivity—precisely the R9 result. This is the opposite of the "clear, crisp, causal" substrate the programme wants.
- **Contradiction with the parallel track.** The causal-fibres/PC-GPT-2 track reports the error field converging to the BP adjoint field (gradient-ratio $\eta \to 0.98$ at cosine ≥ 0.998) with frontier mass early-to-middle (66% in blocks 2–5). R8/R9 reports credit concentrated in the final block. Two tracks of one programme give opposite answers about where credit lives. If R8/R9 ran sPC-with-decay and the other track ran a correct (or effectively error-parameterized) scheme, the contradiction dissolves. This is a testable claim, listed as Prediction 1 below.

Finding 1. *Every headline negative result in the six-layer studies (no loss win, rank collapse, weak selectivity, late-block credit, wall-clock blowup) is a predicted consequence of running state-based PC at shallow settlement depth with a depth-non-neutral energy metric. None of it bears on error-based PC.*

3 Finding 2: the preflight control failed like a bug, not like a hard task

3.1 The control is worse than chance

The final r5 fixture has a 64-token vocabulary, so a uniform predictor scores NLL $= \ln 64 \approx 4.16$ nats per token. The recorded T=1 NLLs range from 6.76 to 11.92 across regimes and seeds (perturbation worst at ≈ 11.6 mean). A model that had learned *nothing* would beat the trained control in every regime. Confident wrongness of this magnitude has two standard causes:

1. **Train/eval mapping mismatch.** The evaluation scores a different mapping than training optimized: an off-by-one in the continuation-position index; a cap-ownership convention that differs between the teacher-writing path and the scoring path; or the nuisance cue leaking into the salt on the perturbation path. The perturbation regime being by far the worst is exactly the signature of the cue perturbing the *target lookup* rather than only the input.
2. **Aggregation miscalibration.** If the reported NLL is summed over the four continuation positions (and/or candidates) rather than averaged per scored token, chance is $4 \ln 64 \approx 16.6$ and the frozen floor of 1.4 was calibrated in different units than the metric, i.e. the gate was unpassable by construction.

Either way this is an instrument defect. Note that the r5 report's own list of five "plausible explanations" contains neither.

3.2 The capacity excuse does not survive arithmetic

The pretraining task is 3 regimes $\times$ 4 contexts $\times$ 2 modes $\times$ 4 positions $= 96$ arbitrary tokens from a 48-token effective range: about $96 \times \log_2 48 \approx 537$ bits of lookup content, trained for 192 updates at batch 32 ($\approx 6{,}100$ examples) in a two-layer, width-32 transformer with tens of thousands of parameters. A model of this size memorizes such a table easily. "The fixture may remain too sparse relative to model width" is not a credible hypothesis; the numbers point at the instrument.

Finding 2. *No chance baseline was computed anywhere in the programme. A single line ("untrained model NLL = . . .; uniform NLL =* $\ln V$*") would have converted r5 from a mysterious construction failure into an immediate bug report, and would have exposed the units of the* 1.4 *floor.*

4 Finding 3: monotone energy is a vacuous invariant

The backtracking line search *accepts a step only if the energy does not increase.* Monotone energy traces are therefore true by construction; they verify that the line search compiles and nothing else. Yet "monotone activity energy, 10/10 seeds" is reported as the programme's most reproducible positive finding across 62 records.

What settlement actually needs to certify is *convergence*:

- the relative residual $\rho_T = \|\nabla_z E(z^{(T)})\| / \|\nabla_z E(z^{(0)})\|$ at the last recorded step;
- steps-to-tolerance for a fixed $\rho^\star$ (e.g. 10^{-2}, 10^{-4}), reported per layer;
- conditioning of the settling map (power-iteration estimate of the spectral radius of its Jacobian at the settled point);
- variance of all of the above across inputs.

If $\rho_T \approx 0.9$ at $T{=}4$—which is what shallow sPC predicts— there is no fixed point in play, the "$T{>}1$ arm" is not a PC arm in any meaningful sense, and every downstream comparison is between BP and BP-plus-noise. The appendix makes these convergence certificates mandatory (Gates MG-3/MG-4).

5 Finding 4: the programme fought a theorem, and skipped its own question

5.1 Matching BP is the ceiling, not the floor

At the PC energy fixed point, in the small-output-weight regime, PC weight gradients converge to the backpropagation gradients (the well-known equilibrium equivalence; the parallel track's measurement of the error field approaching the BP adjoint at cosine ≥ 0.998 is an empirical observation of exactly this). Consequently the question "does ePC beat matched BP on held-out source loss at matched updates?"—asked and answered negatively in the credit gate, R7, R8, and the adaptation battery—has an approximately known answer: *no, by design.* In the regime where ePC is faithful, it reproduces BP; deviations from BP are either un-converged settlement (noise) or deliberately non-BP objectives (which must then be justified on other grounds). Loss parity at matched quality is the correct *constraint*; it can never be the *prize.*

5.2 The actual prize was never operationalized

The stated goal of the conversion programme is a substrate: a transformer re-expressed as an ePC network whose settled states and error units are clear, crisp, causal, and easy to attach further PC models to (caps, sidecars, the multiscale continual-learning cap, the causal-fibres symbolic head). Nothing in the record defines a metric for this property, measures it, or compares ePC-converted and BP bases as *cap substrates*. Instead the programme escalated toward an MoE/Pareto-frontier question that presupposes a mechanism-level effect nobody had demonstrated.

Finding 3. *The one experiment the programme exists to run—fit an identical PC cap on an ePC-converted base and a quality-matched BP base, and compare cap-side sample efficiency, retention, and interference—was never run.*

6 The organizing question: the λ-window

Let λ (more generally, the ratio of output precision to layerwise precisions Π_l) parameterize the energy (1). Two limits bracket the design space:

- $\lambda \to 0$: the settled state is exactly the feedforward pass; ePC adds nothing but cost.
- λ large: settlement departs strongly from the feedforward pass; function is no longer guaranteed, conditioning degrades, and (as R8/R9 illustrate in the sPC case) representation can collapse.

The scientifically prior question for the whole conversion programme is therefore one-dimensional and cheap:

> *Is there a window of λ in which (a) the settled state departs materially from the feedforward pass, (b) the settling fixed point is stable and well-conditioned, and (c) task function is preserved at matched quality?*

If yes, cap-readiness metrics (Section 7) are measured inside that window, and the window's width is itself a headline result. If no, the conversion goal must be redesigned before any frontier, MoE, or scale question is admissible. Nothing in the July–August record measures this.

7 The redesigned programme: M0, C1, C2, C3

7.1 M0: analytic equivalence gates (CPU, hours)

On deep *linear* networks (depths 2/6/12/24) with a quadratic output term, the PC equilibrium is available in closed form and the small-λ weight gradient must match the BP gradient to numerical tolerance ($\leq 10^{-6}$ relative). Run the implementation through this gate at every depth.

Prediction 1. A correct EO/ePC implementation passes M0 at all depths. The current sPC implementation passes at depth 2 and fails with increasing depth via the attenuation profile; its per-layer error field diverges from the BP adjoint in exactly the depth-decaying pattern seen in the 15 July gate. Running M0 on both implementations therefore also adjudicates the contradiction with the adjoint-field track.

M0 would have caught the mislabeling on day one. An "M0 equivalence check" already exists on paper in the causal-fibres v1.1 programme document; it was never executed on this track.

7.2 C1: conversion by homotopy, not distillation from scratch

Do not train ePC students from scratch against teachers. Take an already competent transformer (GPT-2 class, or the existing PC-GPT-2 checkpoints) and *convert* it:

1. Initialize with λ (and precisions) such that the settled state is exactly the feedforward pass; verify bitwise/functional identity.
2. Anneal λ upward along a schedule of rungs. At each rung: settle to certified convergence (MG-3), lightly fine-tune weights under the settled-state objective, and record the C2 metric battery.
3. Stop when function degrades beyond a pre-registered matched-quality band, or when conditioning fails.

This yields a *curve* of substrates indexed by λ, retains a working model at every point, reuses the function-pinned homotopy machinery already specified in the C4-prime document (applied here to the base rather than the critic), and reduces compute from training runs to hundreds of conversion updates.

7.3 C2: cap-readiness metrics, pre-registered

"Crisp, causal, easy to build on" becomes the following battery, measured along the C1 path (full operational definitions in Appendix A.5):

1. **Settling conditioning**: ρ_T, steps-to-tolerance, Jacobian spectral radius, and their variance across inputs.
2. **Injection response**: clamp a constraint at layer l (the elementary act of a cap) and measure the settled-state response operator $\partial z^\star/\partial u$: its effective rank, sparsity, locality, and cross-context stability. A cap can only attach reliably if this operator is low-rank, stable, and approximately transportable across contexts. *This, not raw effective rank, is the headline cap-readiness number.*
3. **Error-unit informativeness**: linear-probe accuracy of ϵ_l for "which factor was violated" under controlled single-factor perturbations.
4. **Perturbation additivity**: responses to two independent perturbations should approximately add; super-additive entanglement means the substrate is not causal in the intended sense.
5. **Rank preservation**: representation rank within a band of the BP base's; collapse to single-digit rank is a disqualifier, not "distinct geometry."

7.4 C3: the falsifier

Fit an *identical* cap—the multiscale continual-learning PC cap, or the causal-fibres sidecar—on (i) an ePC-converted base from inside the λ-window and (ii) a BP base matched on base perplexity. Compare, with frozen promotion rules and disjoint seed splits: cap sample efficiency to target quality; cap continual-learning retention; and cap-to-base interference. This is the experiment the programme is *for*. Only if C3 returns a stable positive signal do frontier, diversity, MoE, or scale questions become admissible.

8 Engineering and fixture directives (summary)

Full prescriptive versions appear in the appendix; headline items:

- Port the reference EO implementation (`cgoemaere/error_based_PC` lineage); do not reimplement from the paper; verify against published results before touching RelaLeap fixtures.
- Learned or empirically-set per-layer precisions Π_l; define the energy in LayerNorm-whitened coordinates so the metric is depth-neutral. A single global λ with τ=1 against raw residual-stream magnitudes guarantees late-block domination.
- Sublayer granularity: attention and MLP as separate PC edges, giving finer error units and cap attachment points.
- Build in FabricPC (graph-of-nodes/edges, muPC scaling, single graph definition for PC-vs-BP comparisons) rather than bespoke CPU-bound PyTorch runners.
- Replace the salted-lookup fixture. Independently salted tables share no structure; "compose" there is a second memorization task. Use a generative task where composition is defined by shared parts (small PCFG or algebraic/relational grammar) with held-out combination splits, and calibrate control learnability on separate seeds before freezing gates.
- Defer all MoE/Pareto questions until after a C3 signal.

9 Process finding: procedural rigor without mechanistic gates

Every gate in the July–August record is *procedural*: bit-exact replay, digest matching, monotone traces, frozen seeds. No gate is *mechanistic*: does the fixed point exist and is it reached; does the implementation satisfy its defining identity on an analytically solvable case; is the evaluation scoring the mapping training optimized; what is chance. Procedural gates certify that whatever happened, happened reproducibly. Mechanistic gates certify that the thing that happened is the thing you meant to study. The remediation is structural: a mandatory mechanism-gate tier (Appendix A.2) that must pass before any performance comparison is *interpreted at all*, with closed-form positive controls where a computed number must match an analytic value—not merely replay exactly.

A second-order lesson for agent-driven research: the agents faithfully optimized the contract they were given. The contract specified reproducibility invariants and promotion gates but never required reading the method paper past its title, never required a chance baseline, and never required an analytic positive control. Contracts for coding agents must therefore include *comprehension obligations* (Appendix A.1) alongside execution obligations.

10 Conclusion

The record does not show that ePC transformers fail; it shows that sPC at shallow settlement, scored by a broken instrument, fails to beat a theorem. The core idea—converting competent transformers into error-parameterized PC networks and using the settled error structure as a substrate for PC caps—remains untested and, in our judgment, well-posed. The M0/C1/C2/C3 programme above tests it directly, cheaply, and in an order in which each stage's failure is informative: M0 failure means the implementation is wrong; an empty λ-window in C1/C2 means the conversion concept needs redesign; a null C3 means ePC substrates offer caps no advantage over BP substrates at matched quality, which would be a genuine, decisive, publishable negative result—unlike anything in the current record.

A Detailed guidance for coding agents

This appendix is normative. "MUST/MUST NOT" are hard requirements; "SHOULD" requires written justification to deviate. It is intended to be pasted into agent contracts for the successor programme.

A.1 Standing rules (apply to every stage)

SR-1. Comprehension obligation. Before implementing a method from a cited paper, the agent MUST produce a half-page method summary in its own words, including: the objective being optimized, the variables being optimized over, and the paper's stated failure modes. For arXiv:2505.20137 specifically: the optimization variables are the *errors* ϵ_l, not the states z_l; state-space relaxation (sPC) is the pathology the paper exists to fix; any implementation that steps on z is NOT an implementation of this paper.

SR-2. Port, don't reimplement. When a reference implementation exists (the EO authors' repository), the agent MUST port and wrap it, and MUST reproduce at least one published result from the reference code before integrating it with RelaLeap fixtures. Reimplementation from the paper is permitted only after the port passes, as a cross-check.

SR-3. Chance baselines are mandatory. Every results table that contains a loss or NLL MUST also contain: (a) the uniform-predictor value in the same units ($\ln V$ per token for vocabulary V, scaled by the same aggregation as the metric); (b) the untrained-model value at the same evaluation. Any trained arm scoring worse than the untrained model MUST halt the experiment and open a bug investigation; such a result MUST NOT be reported as a task-difficulty finding.

SR-4. Units of every threshold. Every frozen numeric gate MUST state its units and aggregation (per token / per position / per sequence / per candidate; nats or bits) and MUST be accompanied by the chance value in the same units, demonstrating the gate is achievable in principle.

SR-5. Analytic positive controls. Every mechanism MUST first be exercised on a case with a closed-form answer, and the computed number MUST match the analytic value to a declared tolerance. Bit-exact replay of an unverified number is not verification.

SR-6. Separation of instrument and science. Instrument validation (does the eval score what training optimized; do teacher writes land; is the metric computed as declared) MUST be a separate, earlier stage with its own tests, frozen before any science seeds are opened.

SR-7. Fail loudly on worse-than-chance, silently on nothing. Assertions comparing every reported metric against its chance value MUST be part of the runner, not the report.

SR-8. Epistemic labels. Preserve the current record's discipline: every claim labeled observed / decided-under-gate / hypothesis. Add a fourth label: *instrument-validated* (the measurement itself passed its analytic control).

A.2 The mechanism-gate tier (MG)

No performance comparison between arms may be interpreted, reported as evidence, or used in promotion decisions until all MG gates pass on the configuration in question. MG gates are standing regression tests: they run in CI on every change to settlement, energy, precision, or fixture code.

MG-1. Parameterization identity. A unit test MUST verify that the optimization variables are ϵ, i.e. that a perturbation applied to the output-layer discrepancy produces a nonzero update to the layer-1 error variable *within the first inference step*, at every tested depth. (Under sPC this is zero for depth > step count; under EO it is nonzero immediately.)

MG-2. Linear-network equivalence (M0 core). On deep linear networks, depths $\{2, 6, 12, 24\}$: at convergence and small output weight, the relative difference between PC and BP weight gradients MUST be $\leq 10^{-6}$ per layer. Record the full depth profile; a depth-decaying profile is an automatic sPC diagnosis.

MG-3. Convergence certificate. Every settlement MUST record $\rho_T = \|\nabla E(z^{(T)})\|/\|\nabla E(z^{(0)})\|$ (computed in the active parameterization). An arm MAY be labeled "settled" only if $\rho_T \leq 10^{-2}$ (report also the 10^{-4} census). Arms not meeting this MUST be labeled "partial-settle(ρ)" in every table. Monotone-energy traces MUST NOT be reported as evidence of anything.

MG-4. Fixed-point conditioning. At the settled point, estimate the spectral radius of the settling-map Jacobian by power iteration (matrix-free, ≤ 20 iterations). Radius ≥ 1 on any input in the eval set fails the gate for that configuration.

MG-5. Teacher-write integrity. Retain and extend the r2 regression test: exactly one candidate attains the maximal teacher logit at every scored position, and its argmax sequence equals the planted target—checked *through the same code path the trainer consumes*, not a parallel reconstruction.

MG-6. Eval–train correspondence. A round-trip test MUST verify that an oracle model constructed to output the teacher distribution achieves NLL $\leq$ (declared floor)/2 under the *evaluation* code on every regime. If the oracle cannot pass the gate, the gate or the eval is broken; no training run may start.

MG-7. Metric-neutrality of the energy. With whitened coordinates / per-layer precisions active (A.4), the per-layer energy shares at the feedforward point MUST be within a factor of 4 of uniform across layers on a calibration batch. Late-block domination beyond this bound fails the gate.

MG-8. Determinism (retained). Bit-exact replay and digest matching as in the current record. These remain necessary; they are simply no longer sufficient.

A.3 Stage M0 specification

Question. Does the implementation satisfy the defining equivalence of ePC on analytically solvable cases?

Setup. Deep linear networks $z_l = W_l z_{l-1}$, depths $\{2, 6, 12, 24\}$, widths $\{16, 64\}$; quadratic output loss to a fixed target; Gaussian inputs; float64.

Protocol.

1. Compute the BP weight gradient in closed form.
2. Run settlement to $\rho \leq 10^{-8}$; compute PC weight gradients at $\lambda \in \{10^{-3}, 10^{-2}, 10^{-1}\}$.

3. Report per-layer relative gradient error and per-layer error-field vs. BP-adjoint cosine, as functions of depth and λ.
4. Run the identical protocol through the *legacy* (sPC) code path and archive both profiles side by side.

Pass. MG-2 at all depths for the new implementation. *Deliverable.* One table, two profile plots, and a one-paragraph adjudication of the R8/R9-vs-adjoint-track contradiction (Prediction 1).

A.4 Stage C1 specification: homotopy conversion

Inputs. A competent pretrained transformer (GPT-2-small class or existing PC-GPT-2 checkpoints); the M0-passed ePC engine; FabricPC graph definition with attention and MLP as separate edges.

Energy definition (normative).

- Errors defined in LayerNorm-whitened coordinates, OR per-edge precisions Π_l set to the inverse empirical residual covariance (diagonal approximation acceptable) estimated once on a calibration set and frozen. MG-7 must pass either way.
- Output term: KL to the model's own pre-conversion logits (function pinning), weight λ on the anneal schedule.

Protocol.

1. Rung 0: λ chosen so the settled state equals the feedforward pass; assert functional identity on a held-out batch (max logit deviation $\leq 10^{-4}$).
2. Anneal λ over ≥ 8 log-spaced rungs. Per rung: settle to MG-3; $\leq$ a few hundred light weight updates under the settled objective; run the full C2 battery; checkpoint.
3. Stop rules: held-out quality leaves the pre-registered matched band (e.g. Δppl $\leq 2\%$), or MG-4 fails.

Deliverable. The λ-curve: every C2 metric and quality metric as a function of λ, with the admissible window (if any) identified. No promotion decisions at this stage; the curve is the result.

A.5 Stage C2 specification: cap-readiness battery (operational definitions)

All metrics computed at each C1 rung, three seeds, disjoint calibration and measurement batches.

1. **Conditioning:** ρ_T; steps to $\rho \leq 10^{-2}$ and 10^{-4}; MG-4 spectral radius; interquartile ranges across 256 inputs.
2. **Injection response.** For layers $l \in \{\text{early, mid, late}\}$: add a clamped offset u ($\|u\| = 5\%$ of typical state norm, 16 random directions) to the layer-l error unit; re-settle; form the response matrix $R_l = \partial z^\star / \partial u$ via finite differences over the direction set. Report: effective rank of R_l (participation ratio); energy fraction of R_l within layers $\{l, \ldots, l{+}1\}$ (locality); and cross-context transport stability $s_l = \text{mean}_{c \neq c'} \, \text{corr}(R_l^{(c)}, R_l^{(c')})$ over 8 contexts. **Headline cap-readiness number:** s_l at mid layers together with $\text{rank}(R_l)$; a substrate is cap-ready only if the response is low-rank, local, and context-stable.

3. **Error informativeness.** Construct controlled single-factor input perturbations (factor set fixed per fixture); train linear probes from ϵ_l to factor identity on calibration data; report held-out probe accuracy per layer. BP baseline: probes on activation differences of the unconverted model.

4. **Additivity.** For perturbation pairs (a, b) applied jointly: report $\|\epsilon(a+b)-\epsilon(a)-\epsilon(b)\|/(\|\epsilon(a)\|+\|\epsilon(b)\|)$, distribution over 64 pairs. Values $\gg 0.3$ at small perturbation scale indicate entanglement.

5. **Rank band.** Entropy effective rank and participation ratio per layer; pre-registered disqualifier if final-layer effective rank falls below $0.5\times$ the unconverted model's.

A.6 Stage C3 specification: the matched-cap falsifier

Arms. (i) ePC base at the best λ inside the C1 window; (ii) BP base matched to (i) on held-out perplexity within the pre-registered band; (iii) unconverted original base (anchor).

Cap. One fixed cap design used identically on all arms (the multiscale PC cap or the causal-fibres sidecar; choose one and freeze it). Base frozen; cap trained under an identical frozen recipe.

Endpoints (frozen before any run).

1. Cap sample efficiency: updates (and examples) to reach a target cap quality; area-under-learning-curve.
2. Continual retention: two-task sequence through the cap; forgetting of task 1 at matched task-2 quality.
3. Interference: base-behavior drift (KL of base logits pre/post cap training) at matched cap quality.

Splits. Exploration / selection / confirmation seed discipline as in the current record. *Interpretation rule.* Any ePC-vs-BP difference is provisional until it survives untouched confirmation seeds; a null result here is a real, reportable negative about ePC substrates (unlike anything in the current record). MoE/frontier/diversity questions are admissible only after a confirmed positive C3 signal.

A.7 Fixture requirements for any synthetic task

FX-1. Compositional structure MUST be real: held-out cells of a context $\times$ mode grid whose targets are *derivable* from trained cells (shared-parts generative process: small PCFG, group action, or relational grammar). Independently salted lookup tables are banned as composition tests—they measure memorization twice.

FX-2. An explicit oracle model MUST exist in code and MUST pass every regime gate under the evaluation path (MG-6) before any training run.

FX-3. Control learnability MUST be calibrated on dedicated calibration seeds, and the calibrated schedule frozen, before science seeds open. Tuning on gate seeds converts the gate into training data (correctly noted in the current record) and voids the contract.

FX-4. Perturbation regimes MUST verify, by construction test, that the nuisance variable does not enter the target-generation path.

FX-5. Report per-regime chance values in every table (SR-3).

A.8 Compute and tooling directives

CT-1. Implement the ePC engine as a FabricPC graph; the BP comparison arm MUST be generated from the *same* graph definition. Bespoke single-file PyTorch runners are permitted only for MG unit tests.

CT-2. Attention and MLP are separate PC edges; the residual stream is the node state; per-edge precisions are first-class parameters.

CT-3. CPU is acceptable for M0 and MG tests only. C1–C3 run on GPU; the current record's pattern of 200-update "training runs" on laptop CPUs MUST NOT recur—undertrained arms are uninterpretable regardless of replay exactness.

CT-4. Warm-start settlement across adjacent λ rungs and across minibatch steps (persistent error states) SHOULD be used and its effect on steps-to-tolerance reported.

A.9 Reporting requirements

Every stage report MUST contain, in order: (1) the question; (2) instrument-validation results (MG table) for the exact code revision; (3) chance and oracle rows in every results table; (4) observed results with convergence certificates attached to every settled arm; (5) decisions under frozen gates; (6) hypotheses, explicitly labeled, including at least one hypothesis under which the *instrument* rather than the method is at fault; (7) the exact reproduction block (commit, config, hashes) in the style of the current record, which is retained unchanged.

A.10 Failure taxonomy (standing regression content)

The following historical failures MUST exist as named regression tests so they cannot recur silently: `r2_advanced_indexing_copy` (teacher writes through an indexing temporary); `worse_than_chance_halt` (SR-7); `summed_vs_mean_nll_units` (SR-4); `spc_masquerade` (MG-1); `late_block_energy_domination` (MG-7); `vacuous_monotone_energy` (MG-3: a deliberately non-converging settlement must be labeled partial, not settled); `adaptation_only_forgetting` (replay mixture present whenever an adaptation phase exists); `gate_seed_tuning` (FX-3: config hash of the frozen schedule checked against the calibration record).